Raj Kumar Nelluri

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Professional Summary

AI / ML Engineer building production-grade intelligent systems combining machine learning, LLMs, and deterministic decision engines. Experienced in designing end-to-end pipelines from data ingestion to deployed inference, with a focus on explainable and auditable AI systems over black-box models. Built and deployed scalable applications on AWS including financial decision engines, enterprise RAG systems, and predictive ML pipelines.

Technical Skills

Programming: Python, SQL
AI / LLM Systems: Retrieval-Augmented Generation (RAG), LLM Applications, Deterministic Decision Systems, AI System Design
Machine Learning: Scikit-learn, XGBoost, TensorFlow
LLM & AI Tools: OpenAI API, LangChain, Prompt Engineering, Embeddings, Vector Databases
Backend & APIs: FastAPI, REST APIs
Cloud & Deployment: AWS (EC2, SageMaker, Lambda, S3), Docker, Redis
Data & Analytics: Pandas, NumPy, Matplotlib, Tableau, Power BI

AI Engineering Projects

AI Financial Research Agent | **Live Demo** | **GitHub** — 2026

FastAPI | GPT-4o | Alpha Vantage | Redis | AWS EC2 | Docker

- Built a production-grade financial analysis system integrating technical indicators, company fundamentals, and news sentiment into a deterministic multi-factor scoring engine
- Designed normalized scoring with dynamic max adjustment, eliminating missing data bias and reducing false SELL signals from ~100% to threshold-based outcomes
- Developed conflict-aware decision logic detecting disagreement across signals and defaulting to HOLD in ambiguous scenarios to avoid false conviction
- Implemented portfolio allocation engine ranking multiple stocks and distributing capital proportionally based on normalized scores
- Architected and deployed a 7-step async pipeline on AWS EC2 with FastAPI and Redis caching (15 min TTL), reducing repeated query latency from ~ 15s to <100ms
- Integrated GPT-4o strictly for explainability (sentiment classification and reasoning), ensuring all decisions remain deterministic and auditable
- Eliminated reliance on LLM-based decision making by constraining GPT-4o to explanation-only role, ensuring fully reproducible and auditable outputs

Enterprise RAG Chatbot – LLM-Powered Document Intelligence | **Live Demo** | **GitHub** — 2026

Python | LangChain | OpenAI | ChromaDB | Docker | AWS EC2

- Designed and deployed a Retrieval-Augmented Generation (RAG) system enabling users to upload and query PDF documents with citation-grounded responses
- Architecture: PDF → Chunking → Embeddings → ChromaDB → MMR Retrieval → LLM
- Built semantic retrieval pipeline using ChromaDB vector database with Maximum Marginal Relevance (MMR) search and reranking to improve answer relevance
- Implemented hallucination mitigation using similarity-threshold filtering and context-grounded prompts
- Containerized application using Docker and deployed on AWS EC2 with persistent vector storage
- Achieved ~ 2 second end-to-end response latency including retrieval and LLM generation

Customer Churn Prediction Pipeline | **GitHub** — 2025

Python | XGBoost | AWS SageMaker | Lambda | CloudWatch

- Built end-to-end ML pipeline on AWS for customer churn prediction including S3 ingestion, Lambda ETL orchestration, and SageMaker model training
- Deployed XGBoost model as SageMaker real-time inference endpoint and implemented automated monitoring with CloudWatch and EventBridge

Additional Projects

Retail Sales Forecasting System | **GitHub** | *Python, XGBoost, Flask* 2025

- Built time-series forecasting model using XGBoost to predict retail demand from 500,000+ transactions
- Engineered lag features, rolling averages, and seasonal indicators improving demand prediction accuracy

Insurance Fraud Detection System | Python, Scikit-learn, AWS Lambda, Kinesis, RDS 2024

- Developed fraud detection pipeline processing 15,000+ insurance claims using Random Forest and Gradient Boosting models
- Implemented real-time data ingestion using AWS Kinesis and serverless processing with Lambda
- Built ETL validation pipeline and stored fraud scores in AWS RDS enabling analysts to query predictions via SQL

Education

Pace University New York City, NY
Master of Science in Computer Science *Sep 2023 – Apr 2025*

Amrita Vishwa Vidyapeetham Bengaluru, India
Bachelor of Technology in Artificial Intelligence *Jul 2019 – Apr 2023*

Certifications

AWS Certified Cloud Practitioner – Amazon Web Services 2025
Adobe Journey Optimizer Foundation – Adobe 2025
Complete Python Developer – Udemy 2023